

WEEK 2-REPORT

Analysis of Digital Governance Adoption using E-Panchayat Data

Topic : Data Collection, Cleaning, and Exploratory Analysis

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1.Introduction

The main objective of Week 2 was to collect, pre-process, validate, and analyze governance-related datasets associated with the E-Panchayat and e-Gram Swaraj systems in India. The focus of the analysis was to understand the adoption of digital governance practices across different states using Panchayat-level indicators.

Through this dataset, the analysis helps provide an overview of how digital technologies and online governance systems are being utilized in rural areas. The pre-processing and analytical steps were performed using Python and pandas in a Jupyter Notebook environment to prepare the dataset for further exploratory analysis and visualization.

2. Dataset Description.

The dataset used for this analysis contains state-wise governance information related to the E-Panchayat and e-Gram Swaraj systems in India. It includes important indicators such as:

- Gram Panchayat Development Plans Uploaded
- Total Number of Village Panchayats
- Total Number of Block Panchayats
- Zilla Panchayat Plans Uploaded
- Panchayats onboarded to digital systems
- Online payment integration

By comparing the uploaded Panchayat plans with the total number of Panchayats, the dataset helps in understanding the adoption of digital governance technologies across different states. The analysis shows that many Panchayat institutions are gradually

moving towards digital platforms and online governance systems.

These insights can be useful for both governmental analysis and public understanding of digital governance implementation in rural areas.

3.Objectives of the Analysis

The main objectives of this analysis were:

- 1)to collect and study governance-related datasets associated with E-Panchayat and e-Gram Swaraj systems.
- 2)to pre-process and clean the dataset for accurate analysis.
- 3)to validate the consistency and reliability of governance indicators.
- 4)to calculate governance adoption metrics using Panchayat-level data.
- 5)to analyze the adoption of digital governance systems across different states.

6)to identify variations in digital governance implementation in rural administration.

The analysis also aimed to understand how digital technologies are being integrated into Panchayat governance systems and how these insights can support future governance-related decision making.

4.Tools and Technologies Used

The pre-processing and analysis of the dataset were performed using Python in a Jupyter Notebook environment. The pandas library was mainly used for data cleaning, validation, feature engineering, and exploratory data analysis.

The following tools and technologies were used during the task:

- 1)Python
- 2)pandas
- 3)Jupyter Notebook / Google Colab/VS CODE.
- 4)CSV datasets
- 5)Exploratory Data Analysis (EDA) techniques

These tools helped in organizing, cleaning, and analyzing the governance dataset efficiently.

5.Data Collection Process

The dataset used for this analysis was collected from publicly available government governance platforms related to E-Panchayat and e-Gram Swaraj systems in India. The data mainly focused on Panchayat-level digital governance indicators across different states.

The collected dataset include information related to:

- 1) Gram Panchayat Development Plans Uploaded
- 2) Block Panchayat Plans Uploaded
- 3) Zilla Panchayat Plans Uploaded
- 4) Panchayats onboarded to digital systems
- 5) online payment integration
- 6) total Panchayat counts

.

The dataset was downloaded in CSV format and imported into a Jupyter Notebook environment using Python and pandas for pre-processing and analysis.

The selection of this dataset was based on its relevance to digital governance adoption, rural administrative digitization, and public governance transparency. The dataset was considered suitable for analysis because it provided measurable governance indicators that could be compared across states.

6.Data Pre-Processing

Several pre-processing techniques were applied to prepare the dataset for analysis. Initially, the dataset structure and column information were inspected using pandas functions such as head() and info().

The pre-processing steps included:

1. Checking missing values
2. Cleaning and organizing column names
3. Standardizing state names
4. Validating duplicate records
5. Converting numerical columns into proper datatypes.

Additional feature engineering was also performed to calculate:

1)total plans uploaded

2)total Panchayat counts

3)adoption percentage of digital governance systems

These pre-processing steps improved the quality and consistency of the dataset before analysis.

7.Data Standardization

Data standardization was performed to maintain consistency in the dataset and improve analytical accuracy. The state names were cleaned and standardized by removing unnecessary spaces and formatting inconsistencies.

Standardization techniques such as string trimming and title formatting were applied to ensure that all categorical values followed a uniform structure. This step helped prevent duplication and inconsistencies during sorting, filtering, and comparative analysis.

8.Missing Value Analysis

Missing value analysis was conducted to identify incomplete or unavailable data within the dataset. The `isnull().sum()` function in pandas was used to inspect missing values across all governance indicators.

During the pre-processing stage, it was observed that some derived analytical columns initially contained missing values because they had not yet been calculated. After feature engineering and pre-processing, the required governance indicators were successfully generated and validated for further analysis.

9.Feature Engineering

Feature engineering was performed to create additional governance indicators from the existing dataset. New analytical columns were generated to improve comparative analysis and measure digital governance adoption more effectively.

The following derived features were created:

- Total Plans Uploaded
- Total Number of Panchayats
- Adoption Percentage of Digital Governance Systems

The adoption percentage was calculated by comparing the total uploaded Panchayat plans with the total number of Panchayats across states. These engineered features helped in ranking and analyzing governance adoption levels more efficiently.

10.Data Validation

Validation techniques were used to ensure that the dataset was logically correct and suitable for analysis. Several checks were performed to identify possible inconsistencies in the governance indicators.

The validation process included:

- checking duplicate rows
- validating missing values
- identifying negative numerical values
- verifying adoption percentage calculations
- checking datatype consistency

No duplicate records, negative values, or logically impossible adoption percentages were found in the dataset. This confirmed that the dataset was reliable for exploratory analysis and future visualization tasks.

11.Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to identify trends and patterns related to digital governance adoption across Indian states.

The analysis included:

- 1) ranking states based on adoption percentage
- 2) identifying high-performing and low-performing states
- 3) generating statistical summaries
- 4) comparing governance indicators across states

The analysis showed noticeable variation in digital governance adoption levels among different states. Some states demonstrated higher adoption percentages, indicating stronger implementation of online Panchayat governance systems.

12.Preliminary Findings

After cleaning and analyzing the E-Panchayat and e-Gram Swaraj dataset, several important observations were identified regarding digital governance adoption across Indian states.

- 1.The adoption percentage of digital governance services differs significantly from state to state. Some states show high participation in uploading Panchayat development plans, while others have lower adoption levels.
- 2.States with higher adoption percentages appear to have better implementation of digital governance systems and stronger usage of online Panchayat platforms.

3. The dataset indicates that digital governance in rural administration is growing, especially through online planning and Panchayat management systems.
4. Some states still show comparatively lower digital adoption, which may be due to differences in infrastructure, awareness, administrative efficiency, or technology accessibility.
5. During pre-processing and validation, no duplicate records, negative values, or logically incorrect adoption percentages were found. This suggests that the dataset is reliable for further analysis and visualization.
6. The calculated adoption percentage helped in comparing governance performance across states and identifying leading and lagging regions in digital governance implementation.

Overall, the dataset provides meaningful insights into the progress of Panchayat-level digital governance in India and forms a strong foundation for further visualization and policy analysis.

13.Conclusion

The Week 2 task successfully involved the collection, pre-processing, validation, and analysis of governance-related datasets associated with E-Panchayat and e-Gram Swaraj systems.

Through pre-processing and exploratory analysis, meaningful governance indicators were generated from the raw dataset. The analysis helped in understanding the adoption of digital governance technologies in Panchayat systems across different states.

The cleaned and validated dataset now provides a good foundation for future visualization, reporting, and policy-related analysis in the upcoming stages of the project.

14.Screenshots and Workflow Documentation

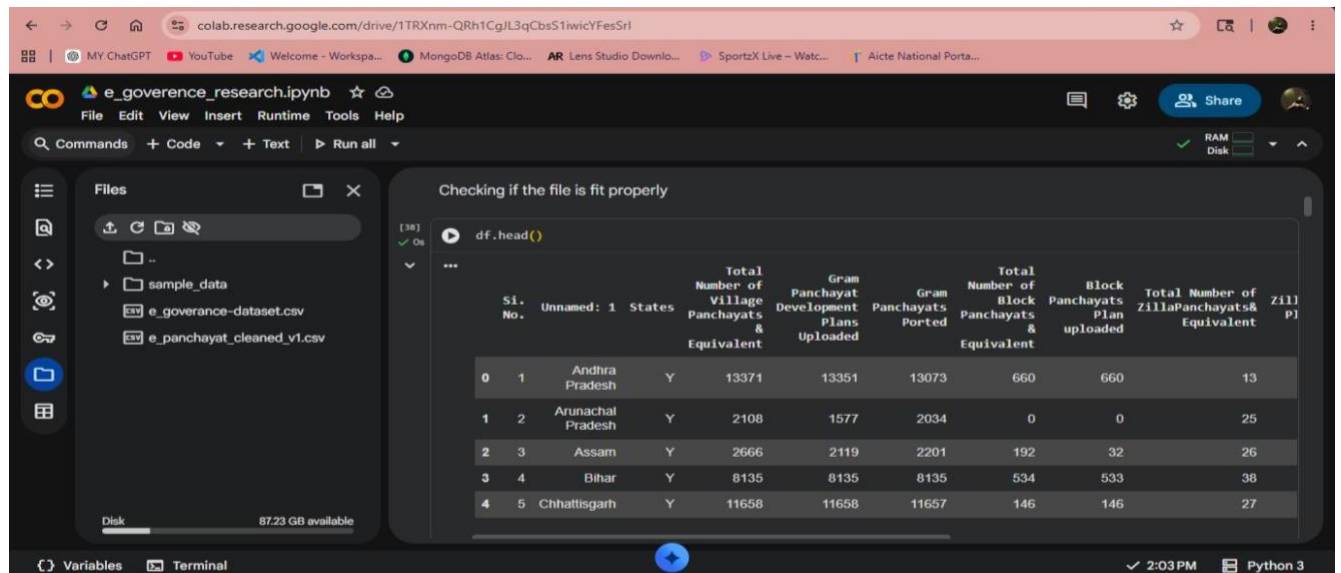
The following screenshots were captured during the pre-processing and analysis workflow:

- Dataset preview using head()
- Dataset structure inspection using info()
- Missing value analysis
- Feature engineering calculations
- Adoption percentage calculation
- Validation checks
- Ranking analysis of states
- Statistical summary generation

These screenshots provide evidence of the pre-processing, validation, and analytical steps performed during the task.

Figure 1: Dataset Preview.

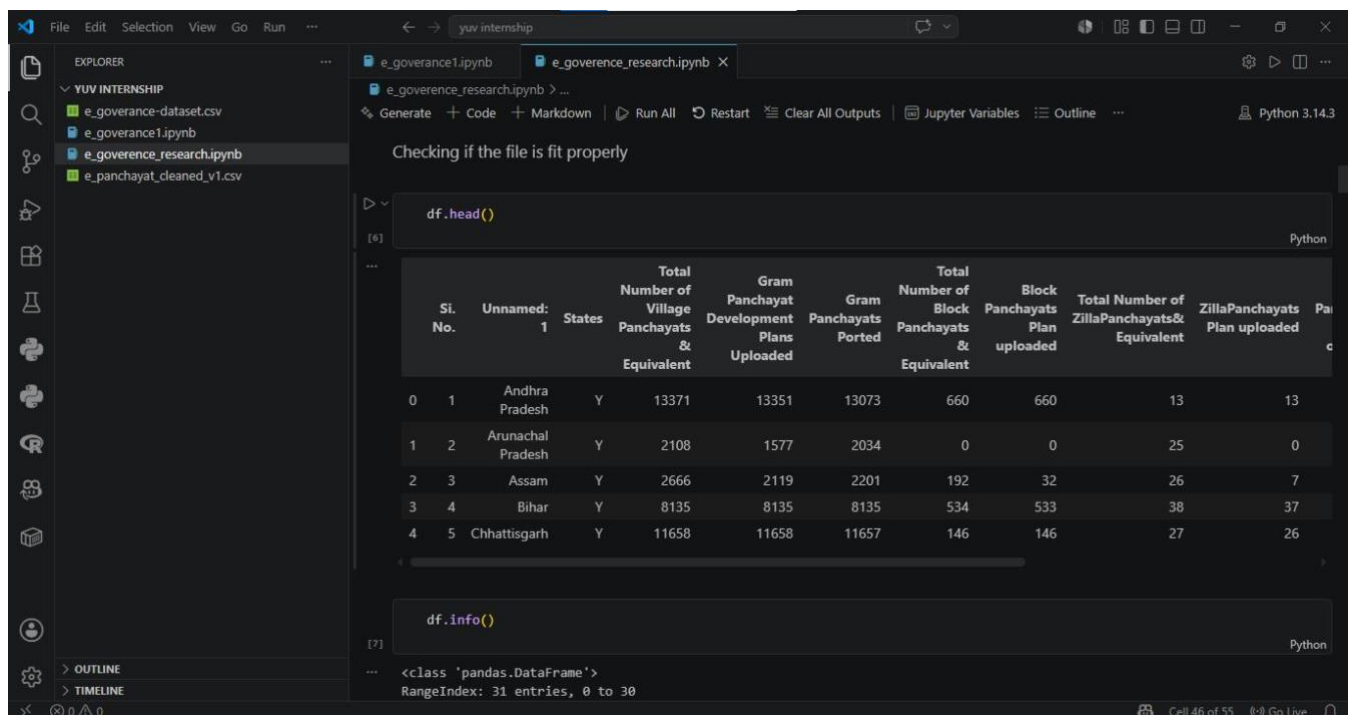
In Google Colab:



The screenshot shows the Google Colab interface with a notebook named 'e_governance_research.ipynb'. The file explorer on the left shows a folder 'sample_data' containing 'e_governance-dataset.csv' and 'e_panchayat_cleaned_v1.csv'. The code cell shows 'df.head()' and the output is a table preview of the dataset.

Si. No.	Unnamed: 1	States	Total Number of Village Panchayats & Equivalent	Gram Panchayat Development Plans Uploaded	Gram Panchayats Ported	Total Number of Block Panchayats & Equivalent	Block Panchayats Plan uploaded	Total Number of Zilla Panchayats & Equivalent	Zilla Panchayats Plan uploaded
0	1	Andhra Pradesh	Y	13371	13351	13073	660	660	13
1	2	Arunachal Pradesh	Y	2108	1577	2034	0	0	25
2	3	Assam	Y	2666	2119	2201	192	32	26
3	4	Bihar	Y	8135	8135	8135	534	533	38
4	5	Chhattisgarh	Y	11658	11658	11657	146	146	27

In VS Code:



The screenshot shows the VS Code interface with a notebook named 'e_governance_research.ipynb'. The Explorer panel on the left shows a folder 'YUV INTERNSHIP' containing 'e_governance-dataset.csv', 'e_governance1.ipynb', 'e_governance_research.ipynb', and 'e_panchayat_cleaned_v1.csv'. The code cell shows 'df.head()' and the output is a table preview of the dataset. Below the table, the 'df.info()' output is shown.

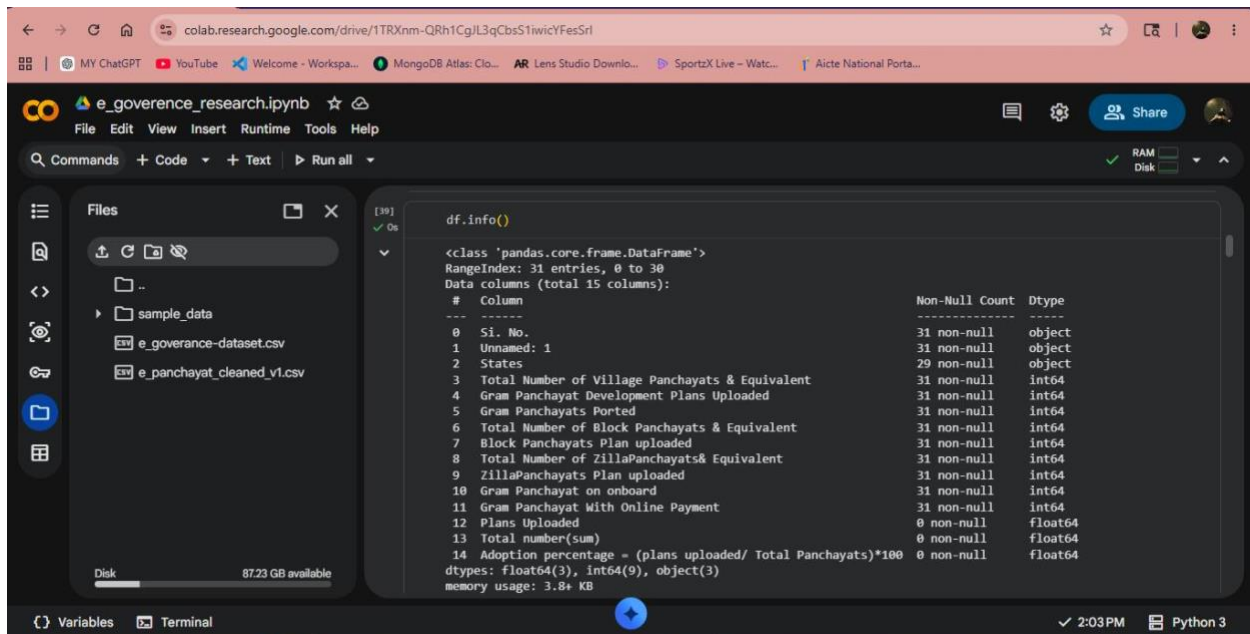
Si. No.	Unnamed: 1	States	Total Number of Village Panchayats & Equivalent	Gram Panchayat Development Plans Uploaded	Gram Panchayats Ported	Total Number of Block Panchayats & Equivalent	Block Panchayats Plan uploaded	Total Number of Zilla Panchayats & Equivalent	Zilla Panchayats Plan uploaded
0	1	Andhra Pradesh	Y	13371	13351	13073	660	660	13
1	2	Arunachal Pradesh	Y	2108	1577	2034	0	0	25
2	3	Assam	Y	2666	2119	2201	192	32	26
3	4	Bihar	Y	8135	8135	8135	534	533	38
4	5	Chhattisgarh	Y	11658	11658	11657	146	146	27

df.info()

```
<class 'pandas.DataFrame'>  
RangeIndex: 31 entries, 0 to 30
```

Figure 2: Dataset structure.

In Google Colab :

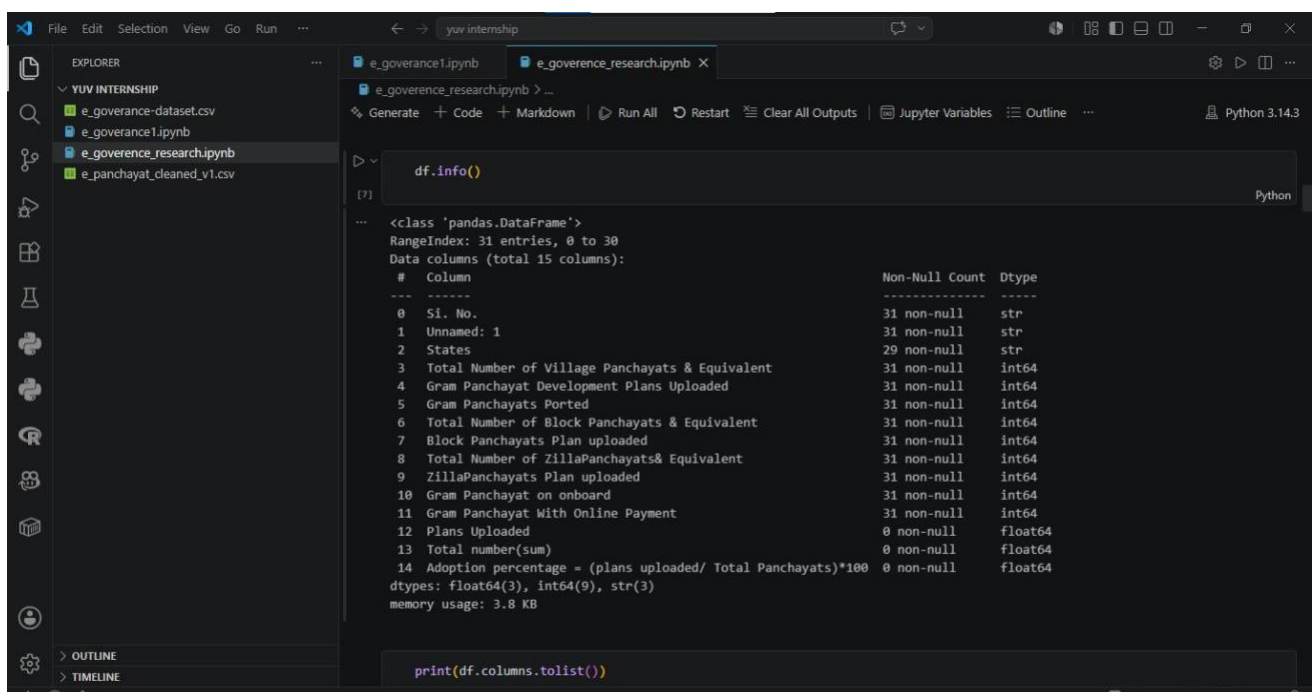


The screenshot shows the Google Colab interface with a file named 'e_governance_research.ipynb'. The left sidebar shows a file explorer with 'sample_data' containing 'e_governance-dataset.csv' and 'e_panchayat_cleaned_v1.csv'. The main code cell shows the output of 'df.info()', which displays the structure of the dataset.

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 31 entries, 0 to 30
Data columns (total 15 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   Si. No.                                                                31 non-null    object
1   Unnamed: 1                                                            31 non-null    object
2   States                                                                29 non-null    object
3   Total Number of Village Panchayats & Equivalent                    31 non-null    int64
4   Gram Panchayat Development Plans Uploaded                        31 non-null    int64
5   Gram Panchayats Ported                                             31 non-null    int64
6   Total Number of Block Panchayats & Equivalent                    31 non-null    int64
7   Block Panchayats Plan uploaded                                    31 non-null    int64
8   Total Number of ZillaPanchayats& Equivalent                    31 non-null    int64
9   ZillaPanchayats Plan uploaded                                    31 non-null    int64
10  Gram Panchayat on onboard                                          31 non-null    int64
11  Gram Panchayat With Online Payment                                31 non-null    int64
12  Plans Uploaded                                                      0 non-null     float64
13  Total number(sum)                                                  0 non-null     float64
14  Adoption percentage = (plans uploaded/ Total Panchayats)*100    0 non-null     float64
dtypes: float64(3), int64(9), object(3)
memory usage: 3.8+ KB
```

In VS CODE:



The screenshot shows the VS Code interface with a file named 'e_governance_research.ipynb'. The left sidebar shows a file explorer with 'YUV INTERNSHIP' containing 'e_governance-dataset.csv', 'e_governance1.ipynb', 'e_governance_research.ipynb', and 'e_panchayat_cleaned_v1.csv'. The main code cell shows the output of 'df.info()', which displays the structure of the dataset.

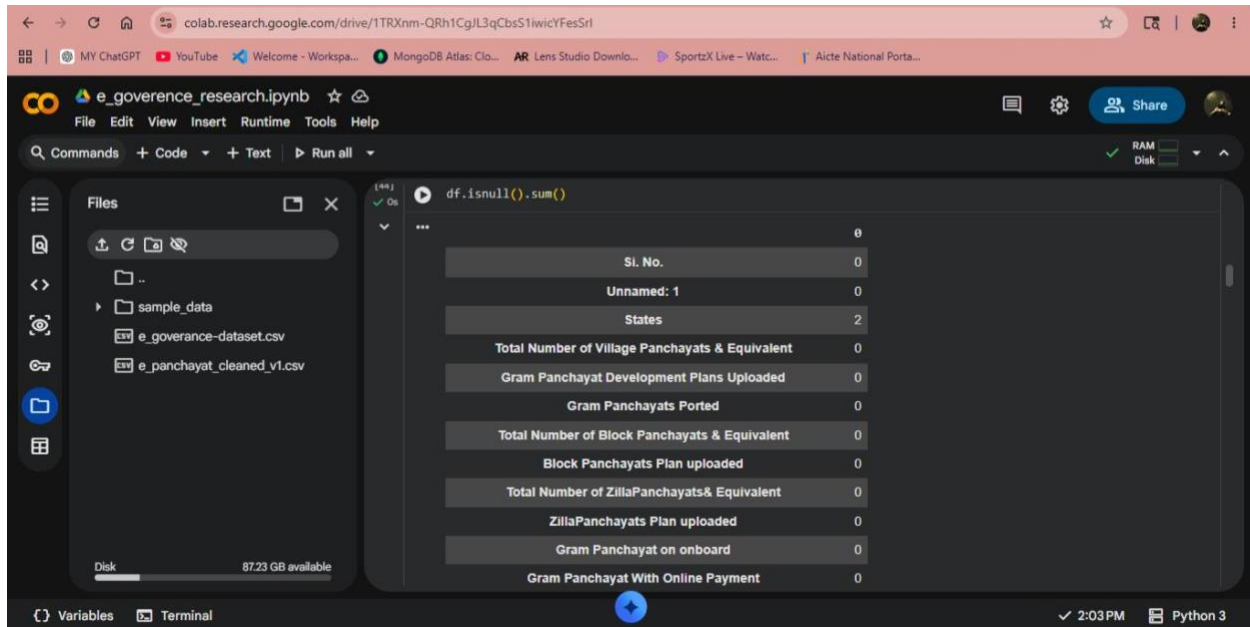
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 31 entries, 0 to 30
Data columns (total 15 columns):
#   Column                                                                 Non-Null Count  Dtype
---  -
0   Si. No.                                                                31 non-null    str
1   Unnamed: 1                                                            31 non-null    str
2   States                                                                29 non-null    str
3   Total Number of Village Panchayats & Equivalent                    31 non-null    int64
4   Gram Panchayat Development Plans Uploaded                        31 non-null    int64
5   Gram Panchayats Ported                                             31 non-null    int64
6   Total Number of Block Panchayats & Equivalent                    31 non-null    int64
7   Block Panchayats Plan uploaded                                    31 non-null    int64
8   Total Number of ZillaPanchayats& Equivalent                    31 non-null    int64
9   ZillaPanchayats Plan uploaded                                    31 non-null    int64
10  Gram Panchayat on onboard                                          31 non-null    int64
11  Gram Panchayat With Online Payment                                31 non-null    int64
12  Plans Uploaded                                                      0 non-null     float64
13  Total number(sum)                                                  0 non-null     float64
14  Adoption percentage = (plans uploaded/ Total Panchayats)*100    0 non-null     float64
dtypes: float64(3), int64(9), str(3)
memory usage: 3.8 KB
```

```
print(df.columns.tolist())
```


Figure 3: Missing Value Analysis.

In Google Colab:



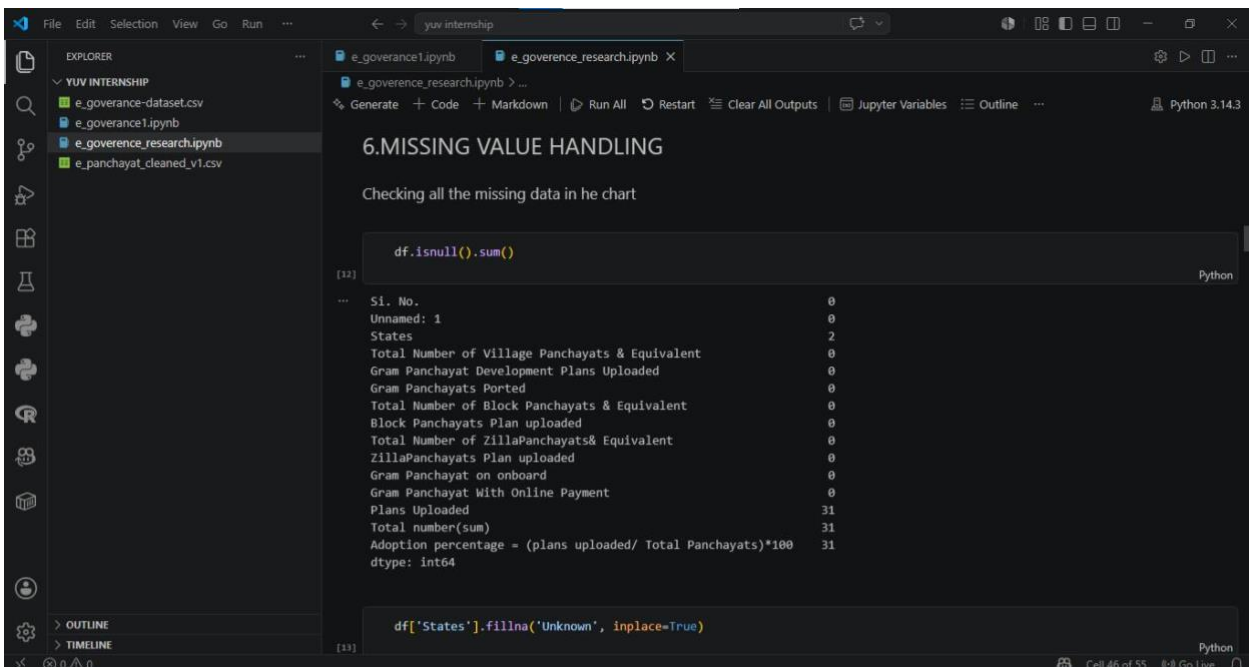
The screenshot shows the Google Colab interface. The file explorer on the left shows a folder named 'sample_data' containing two CSV files: 'e_governance-dataset.csv' and 'e_panchayat_cleaned_v1.csv'. The main code cell contains the following Python code:

```
df.isnull().sum()
```

The output of the code is a table showing the count of missing values for each column:

SI. No.	0
Unnamed: 1	0
States	2
Total Number of Village Panchayats & Equivalent	0
Gram Panchayat Development Plans Uploaded	0
Gram Panchayats Ported	0
Total Number of Block Panchayats & Equivalent	0
Block Panchayats Plan uploaded	0
Total Number of ZillaPanchayats & Equivalent	0
ZillaPanchayats Plan uploaded	0
Gram Panchayat on onboard	0
Gram Panchayat With Online Payment	0

In VS CODE:



The screenshot shows the VS Code interface with a Jupyter Notebook. The Explorer panel on the left shows a folder named 'YUV INTERNSHIP' containing four files: 'e_governance-dataset.csv', 'e_governance1.ipynb', 'e_governance_research.ipynb', and 'e_panchayat_cleaned_v1.csv'. The main code cell contains the following Python code:

```
df.isnull().sum()
```

The output of the code is a table showing the count of missing values for each column:

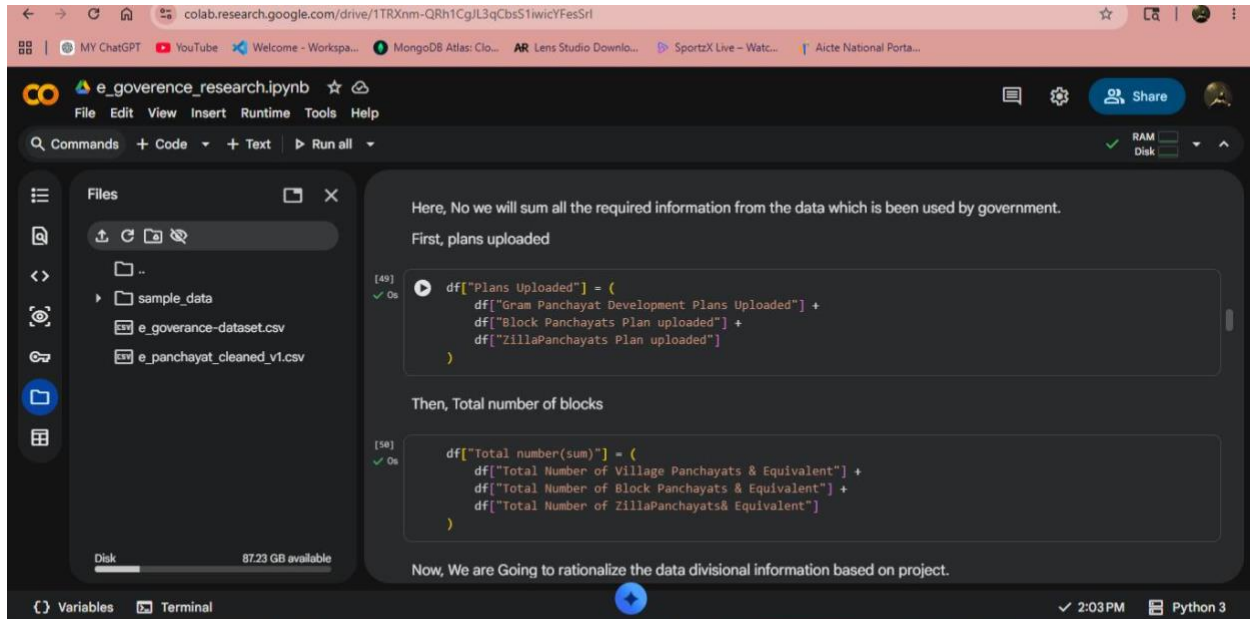
SI. No.	0
Unnamed: 1	0
States	2
Total Number of Village Panchayats & Equivalent	0
Gram Panchayat Development Plans Uploaded	0
Gram Panchayats Ported	0
Total Number of Block Panchayats & Equivalent	0
Block Panchayats Plan uploaded	0
Total Number of ZillaPanchayats & Equivalent	0
ZillaPanchayats Plan uploaded	0
Gram Panchayat on onboard	0
Gram Panchayat With Online Payment	0
Plans Uploaded	31
Total number(sum)	31
Adoption percentage = (plans uploaded/ Total Panchayats)*100	31
dtype: int64	

The next code cell contains the following Python code:

```
df['States'].fillna('Unknown', inplace=True)
```

Figure 4: Feature Engineering

In Google Colab:

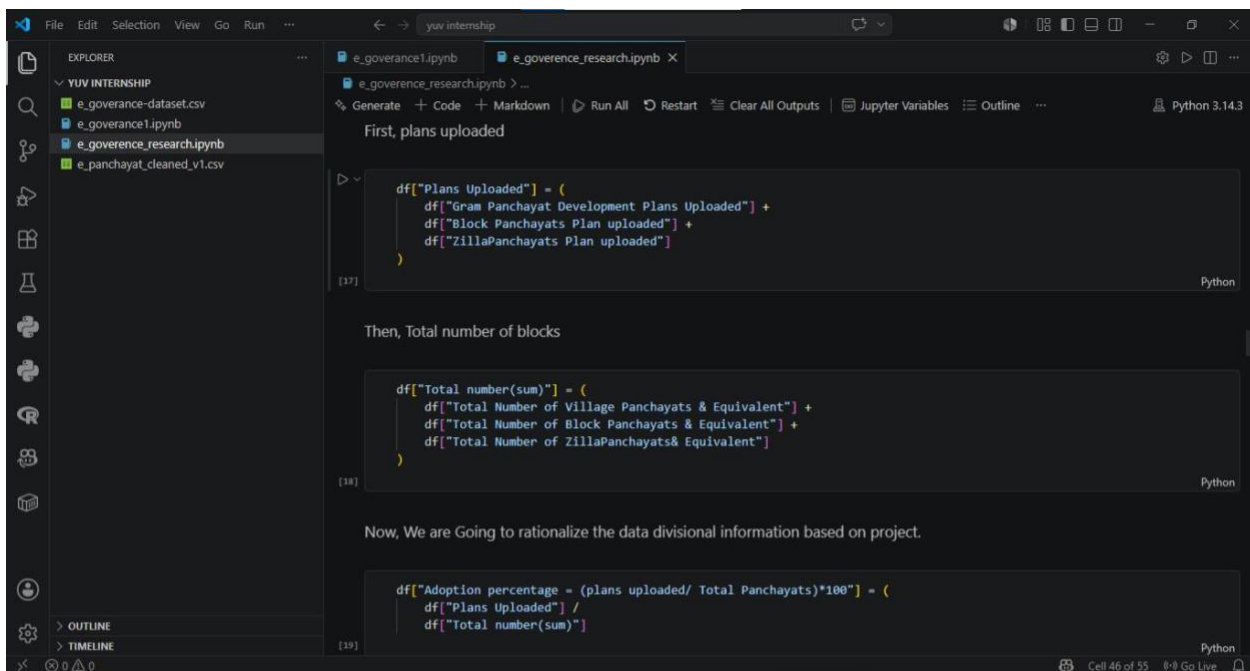


The screenshot shows the Google Colab interface with a notebook titled 'e_governance_research.ipynb'. The left sidebar displays the file explorer with a folder 'sample_data' containing 'e_governance-dataset.csv' and 'e_panchayat_cleaned_v1.csv'. The main editor area contains the following code:

```
[49] df["Plans Uploaded"] = (  
    df["Gram Panchayat Development Plans Uploaded"] +  
    df["Block Panchayats Plan uploaded"] +  
    df["ZillaPanchayats Plan uploaded"]  
)  
  
[50] df["Total number(sum)"] = (  
    df["Total Number of Village Panchayats & Equivalent"] +  
    df["Total Number of Block Panchayats & Equivalent"] +  
    df["Total Number of ZillaPanchayats & Equivalent"]  
)
```

Below the code, there is a text box stating: "Now, We are Going to rationalize the data divisional information based on project."

In VS Code:



The screenshot shows the VS Code interface with a notebook titled 'e_governance_research.ipynb'. The left sidebar displays the file explorer with a folder 'YUV INTERNSHIP' containing 'e_governance-dataset.csv', 'e_governance.ipynb', 'e_governance_research.ipynb', and 'e_panchayat_cleaned_v1.csv'. The main editor area contains the following code:

```
df["Plans Uploaded"] = (  
    df["Gram Panchayat Development Plans Uploaded"] +  
    df["Block Panchayats Plan uploaded"] +  
    df["ZillaPanchayats Plan uploaded"]  
)  
  
df["Total number(sum)"] = (  
    df["Total Number of Village Panchayats & Equivalent"] +  
    df["Total Number of Block Panchayats & Equivalent"] +  
    df["Total Number of ZillaPanchayats & Equivalent"]  
)  
  
df["Adoption percentage = (plans uploaded/ Total Panchayats)*100"] = (  
    df["Plans Uploaded"] /  
    df["Total number(sum)"]
```

Figure 5: Validation Checks

In Google Colab:

The screenshot shows a Google Colab notebook titled 'e_governance_research.ipynb'. The left sidebar displays the file explorer with a folder named 'sample_data' containing two CSV files: 'e_governance-dataset.csv' and 'e_panchayat_cleaned_v1.csv'. The main code area shows two code cells. The first cell, labeled [52], runs `df["States"].nunique()` and outputs the value 3. The second cell, labeled [53], runs a loop to check for impossible adoption using the formula $\text{Adoption percentage} = (\text{plans uploaded} / \text{Total Panchayats}) * 100$. The output of this cell is a table with columns: Si. No., Unnamed: 1, States, Total Number of Village Panchayats & Equivalent, Gram Panchayat Development Plans Uploaded, Gram Panchayats Ported, Total Number of Block Panchayats & Equivalent, Block Panchayats Plan uploaded, Total Number of Zilla Panchayats & Equivalent, and Zilla Pa Plan. The bottom status bar indicates '2:03 PM' and 'Python 3'.

```
[52] df["States"].nunique()
3

checking for the impossible adoption

[53] df[
    df["Adoption percentage" = (plans uploaded/ Total Panchayats)*100"] > 100
]
...
```

Si. No.	Unnamed: 1	States	Total Number of Village Panchayats & Equivalent	Gram Panchayat Development Plans Uploaded	Gram Panchayats Ported	Total Number of Block Panchayats & Equivalent	Block Panchayats Plan uploaded	Total Number of Zilla Panchayats & Equivalent	Zilla Pa Plan

checking the negative values in the dataset.

The screenshot shows the same Google Colab notebook at a later stage. The code area now shows two more code cells. The third cell, labeled [54], runs `negative_values = df[(df.select_dtypes(include='number') < 0).any(axis=1)]` to identify negative values. The output is a table with the same columns as the previous one. The fourth cell, labeled [55], runs `df[["States",` to verify the calculated columns. The bottom status bar indicates '2:03 PM' and 'Python 3'.

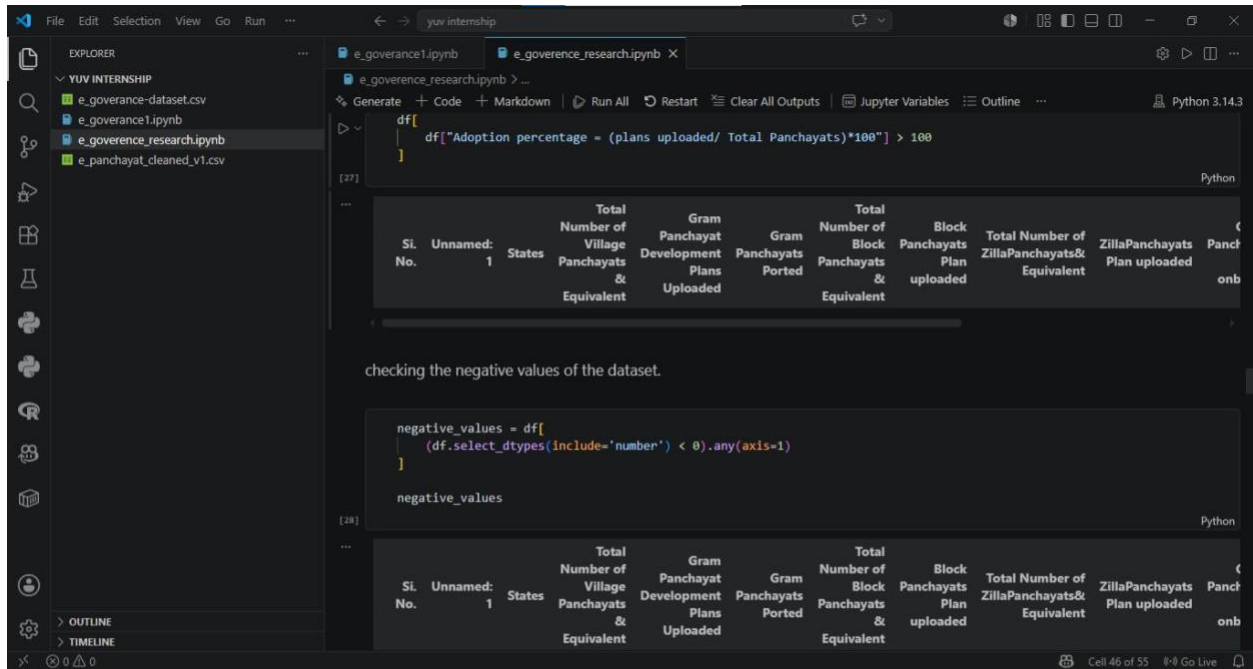
```
[54] negative_values = df[
    (df.select_dtypes(include='number') < 0).any(axis=1)
]
negative_values

[55] df[["States",
```

Si. No.	Unnamed: 1	States	Total Number of Village Panchayats & Equivalent	Gram Panchayat Development Plans Uploaded	Gram Panchayats Ported	Total Number of Block Panchayats & Equivalent	Block Panchayats Plan uploaded	Total Number of Zilla Panchayats & Equivalent	Zilla Pa Plan

verifying the calculated columns

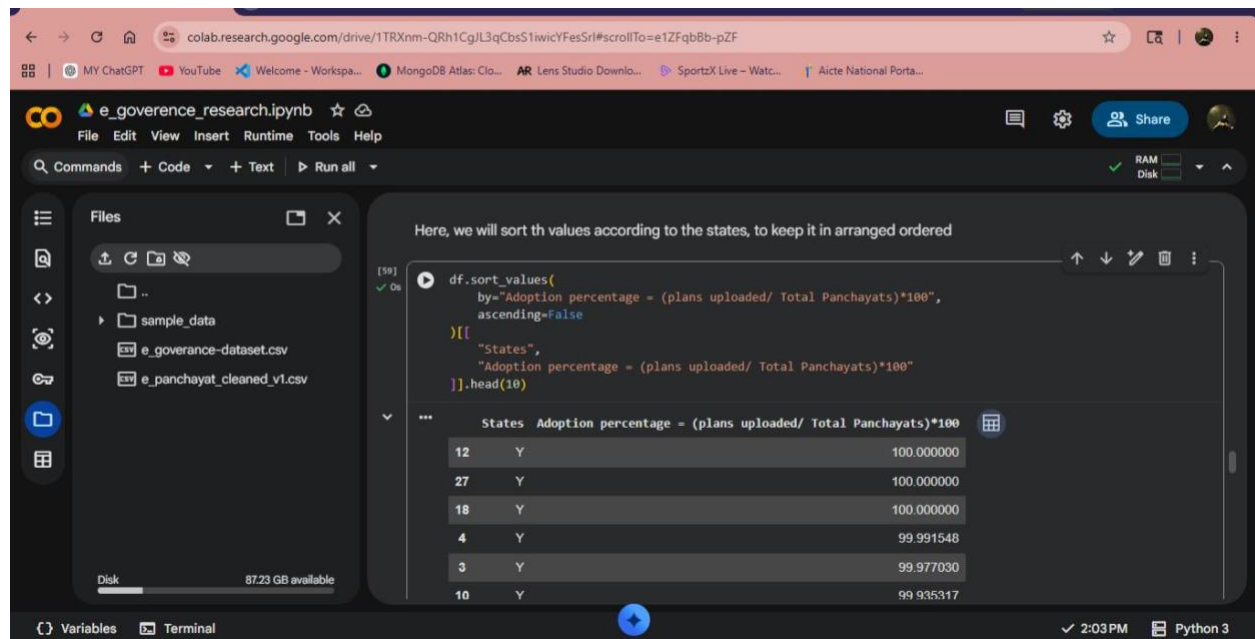
In VS CODE:



(Scroll below)

Figure 5: Ranking Analysis

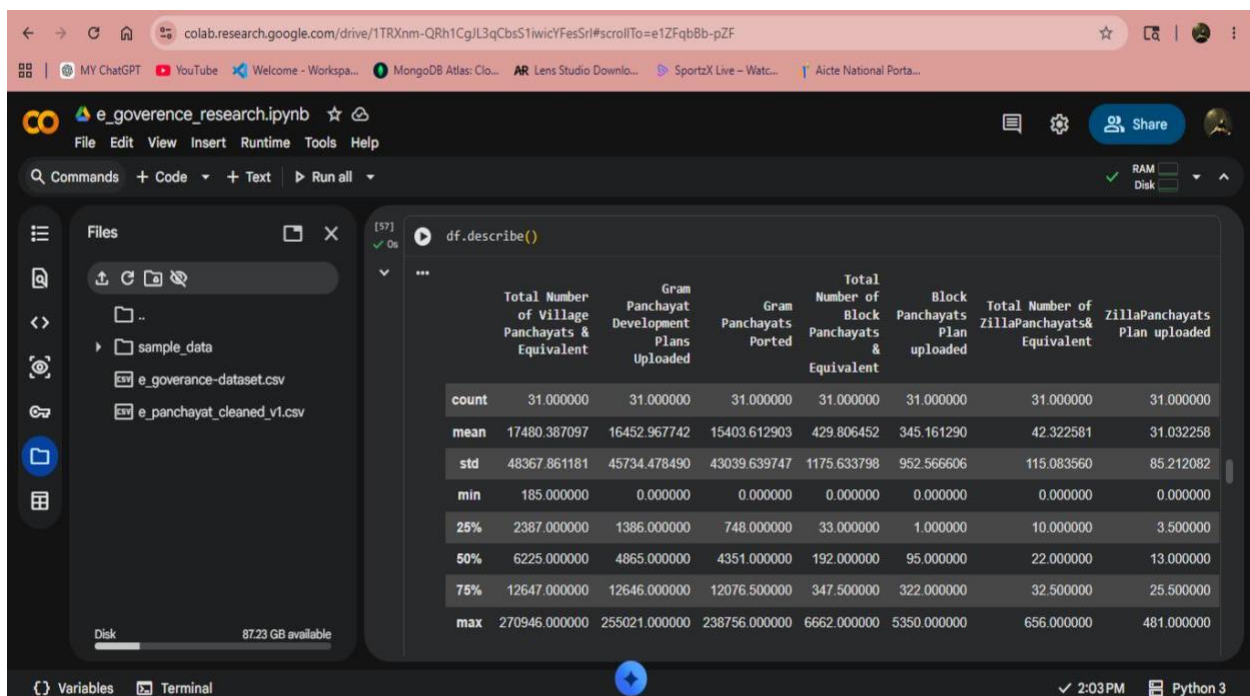
In Google Colab:



The screenshot shows a Google Colab notebook titled "e_governance_research.ipynb". The left sidebar displays the file explorer with a folder named "sample_data" and two CSV files: "e_governance-dataset.csv" and "e_panchayat_cleaned_v1.csv". The main code cell contains a Python snippet that sorts a DataFrame by the "Adoption percentage" column in descending order. The output shows the top 10 rows of the sorted data.

```
[59] df.sort_values(
      by="Adoption percentage = (plans uploaded/ Total Panchayats)*100",
      ascending=False
    )[[
      "States",
      "Adoption percentage = (plans uploaded/ Total Panchayats)*100"
    ]].head(10)
```

States	Adoption percentage = (plans uploaded/ Total Panchayats)*100
12	100.000000
27	100.000000
18	100.000000
4	99.991548
3	99.977030
10	99.935317



The screenshot shows the same Google Colab notebook. The code cell now contains a call to the "describe()" method on the DataFrame. The output displays a summary of the data, including counts, means, standard deviations, and percentiles for various columns.

```
[57] df.describe()
```

	Total Number of Village Panchayats & Equivalent	Gram Panchayat Development Plans Uploaded	Gram Panchayats Ported	Total Number of Block Panchayats & Equivalent	Block Panchayats Plan uploaded	Total Number of Zilla Panchayats & Equivalent	Zilla Panchayats Plan uploaded
count	31.000000	31.000000	31.000000	31.000000	31.000000	31.000000	31.000000
mean	17480.387097	16452.967742	15403.612903	429.806452	345.161290	42.322581	31.032258
std	48367.861181	45734.478490	43039.639747	1175.633798	952.566606	115.083560	85.212082
min	185.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	2387.000000	1386.000000	748.000000	33.000000	1.000000	10.000000	3.500000
50%	6225.000000	4865.000000	4351.000000	192.000000	95.000000	22.000000	13.000000
75%	12647.000000	12646.000000	12076.500000	347.500000	322.000000	32.500000	25.500000
max	270946.000000	255021.000000	238756.000000	6662.000000	5350.000000	656.000000	481.000000

In VS CODE:

Here, we will sort the values according to the states, to keep it in arranged order

```
df.sort_values(
    by="Adoption percentage = (plans uploaded/ Total Panchayats)*100",
    ascending=False
)[[
    "States",
    "Adoption percentage = (plans uploaded/ Total Panchayats)*100"
]].head(10)
```

States	Adoption percentage = (plans uploaded/ Total Panchayats)*100
18	100.000000
27	100.000000
12	100.000000
4	99.991548
3	99.977030
10	99.935317
25	99.932539
24	99.876448
0	99.857590
28	99.854459

So, Describe call is used here to check whether the data formatted is perfectly fit on its block, and to evaluate the data-source

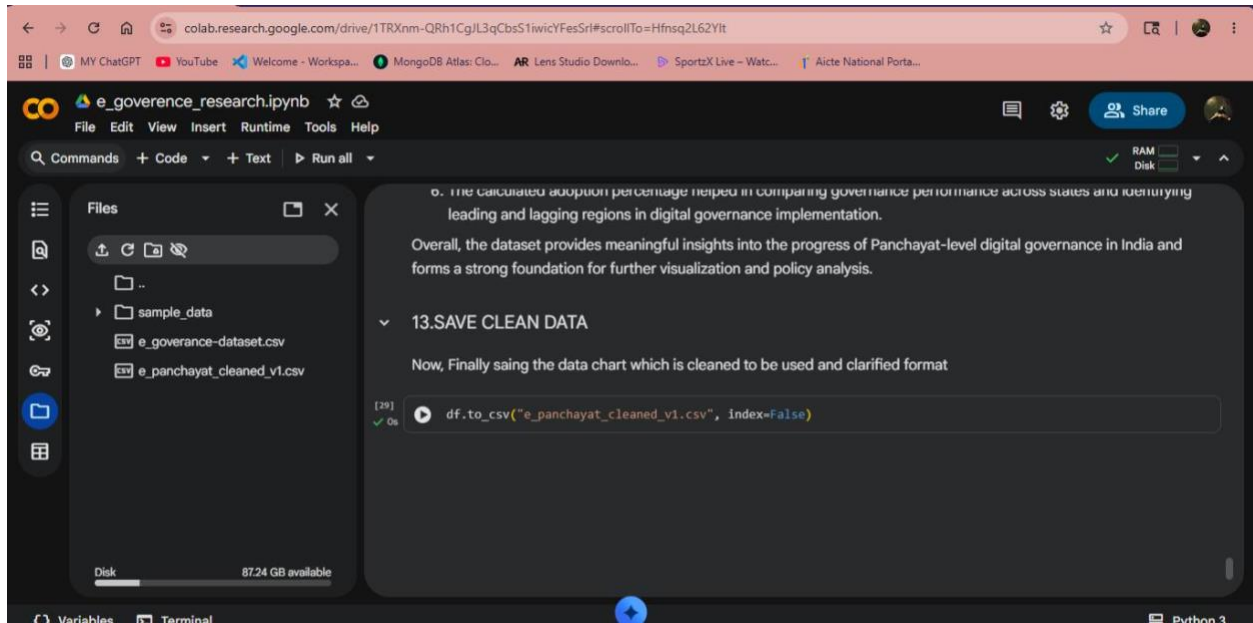
```
df.describe()
```

	Total Number of Village Panchayats & Equivalent	Gram Panchayat Development Plans Uploaded	Gram Panchayats Ported	Total Number of Block Panchayats & Equivalent	Block Panchayats Plan uploaded	Total Number of Zilla Panchayats & Equivalent	Zilla Panchayats Plan uploaded	Gram Panchayat on onboard
count	31.000000	31.000000	31.000000	31.000000	31.000000	31.000000	31.000000	31.000000
mean	17480.387097	16452.967742	15403.612903	429.806452	345.161290	42.322581	31.032258	14152.322581
std	48367.861181	45734.478490	43039.639747	1175.633798	952.566606	115.083560	85.212082	39815.039326
min	185.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	2387.000000	1386.000000	748.000000	33.000000	1.000000	10.000000	3.500000	166.500000
50%	6225.000000	4865.000000	4351.000000	192.000000	95.000000	22.000000	13.000000	3540.000000
75%	12647.000000	12646.000000	12076.500000	347.500000	322.000000	32.500000	25.500000	10837.000000
max	270946.000000	255021.000000	238756.000000	6662.000000	5350.000000	656.000000	481.000000	219361.000000

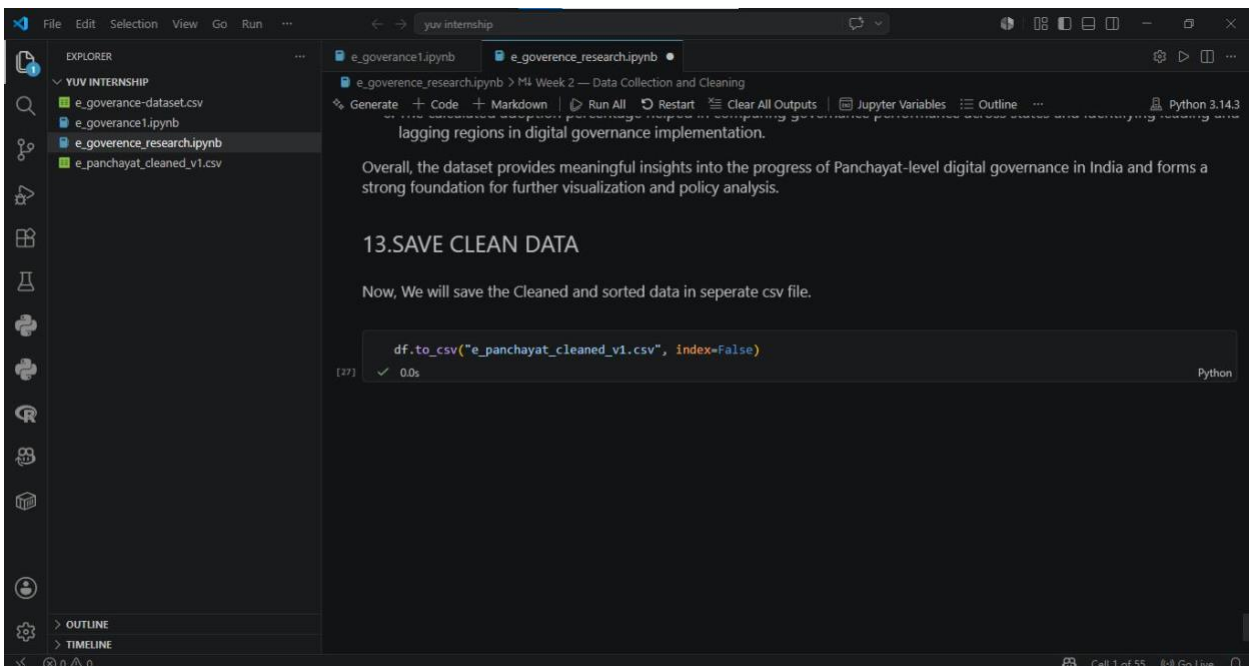
no divisional errors

Figure 6: Statistical summary generation.

In Google Colab:



In VS CODE:



15.References

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<https://egramswaraj.gov.in/>

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<https://panchayat.gov.in/>

3. Python Documentation

<https://www.python.org/>

4. Pandas Documentation

<https://pandas.pydata.org/docs/>

5. Jupyter Notebook Documentation

<https://jupyter.org/>

6. India Open Government data:

<https://www.data.gov.in/>

7. GitHub – E governance & Digital Service:

[Rohan's GitHub Link](#)